Example 2. 1

The average monthly sales, based on past experience of a particular brand of tooth paste in departmental stores is ₹ 200. An advertisement campaign was made by the company and then a sample of 26 departmental stores was taken at random and found that the average sales of the particular brand of tooth paste is ₹ 216 with a standard deviation of ₹ 8. Does the campaign have helped in promoting the sales of a particular brand of tooth paste?

Solution:

Step 1 : Hypotheses

Null Hypothesis H_0: $\mu = 200$

i.e., the average monthly sales of a particular brand of tooth paste is not significantly different from ₹ 200.

Alternative Hypothesis H_1: $\mu > 200$

i.e., the average monthly sales of a particular brand of tooth paste are significantly different from ₹ 200. It is one-sided (right) alternative hypothesis.

Step 2 : Data

The given sample information are:

Size of the sample $(n) = 26$. Hence, it is a small sample.

Sample mean $(\bar{x}) = 216$, Standard deviation of the sample = 8.

Step 3 : Level of significance

$\alpha = 5\%$

Step 4 : Test statistic

The test statistic under H_0 is $T = \dfrac{\overline{X} - \mu_0}{S/\sqrt{n}}$

Since n is small, the sampling distribution of T is the t-distribution with $(n-1)$ degrees of freedom.

Step 5 : Calculation of test statistic

The value of T for the given sample information is calculated from

$$t_0 = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \text{ as}$$

$$t_0 = \frac{216 - 200}{8/\sqrt{26}} = 10.20$$

Step 6 : Critical value

Since H_1 is one-sided (right) alternative hypothesis, the critical value at α =0.05 is

$$t_e = t_{n-1,\,\alpha} = t_{25,0.05} = 1.708$$

Step 7 : Decision

Since it is right-tailed test, elements of critical region are defined by the rejection rule $t_0 > t_e = t_{n-1,\,\alpha} = t_{25,0.05} = 1.708$. For the given sample information $t_0 = 10.20 > t_e = 1.708$. It indicates that given sample contains sufficient evidence to reject H_0. Hence, the campaign has helped in promoting the increase in sales of a particular brand of tooth paste.

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